

Essential Business Deliverable #4: Product Technology Validation

Questions	Answers
State clearly where in the development cycle your product is. 1. Use DOD/DOE TRLs/MRLs to designate the stage of development. 2. Tell us whether an initial prototype or full-scale prototype has been successfully developed. 3. Do you have any initial users/field installations? 4. Is the product ready for manufacture?	1. TRL 3 – We have built and tested a proof-of-concept software model that simulates SPARK’s AI coordination engine using real building energy data. MRL 2 – We have defined conceptual hardware design parameters but have not yet built a prototype. 2. No physical or full-scale prototype has been developed. Software-only prototype is currently under development. 3. No initial users or field installations to date. We are in exploratory discussions with potential pilot customers. 4. The product is not yet ready for manufacturing. Hardware prototyping and compliance planning will begin following completion of the software stack.
How has the underlying technology in your product been validated? 1. Published research articles 2. Internal / pending research that is not yet public - if so, can that be confirmed by third parties? What vetting processes has that research gone through? 3. Patents: have any been published yet, if not, at what stage of application are they? 4. Other products (from your company or other companies) using the same underlying technology?	1. While SPARK has not yet published formal articles, we have submitted technical papers to DOE’s ARPA-E for technical validation. 2. Internal research and software testing have been conducted using real-world commercial building interval data to simulate savings, preference-based control, and override logic. 3. We are in the process of filling a provisional patent for SPARK’s system architecture, coordination logic, and modular panel interface design. PCT filing is targeted for Q1 2026. 4. No other products combine these elements in the same way. However, elements of the technology (load disaggregation, DER coordination, AI-based optimization) have

	been validated independently in other applications.
Has your product (as distinct from the technology) been validated by any third parties (e.g., national labs, universities, research institutions, certification organizations, pilot customers)?	DOE's ARPA-E has validated the technical merit of SPARK's technology. No other third-party product validation has occurred to date. SPARK is in early discussions with potential pilot customers and academic partners but has not yet begun formal validation or testing partnerships.
If so, what stage are these validations in (preliminary talks, pending, completed)?	We have had detailed discussions with DOE and have submitted two technical papers and subsequent comparisons with existing DER integrations technologies. All other third-party engagement is currently at the preliminary talks stage. Formal pilot programs and evaluation partnerships will begin following the development of a working hardware prototype.
Are there any safety or industry standards testing (e.g., UL, ASTM, etc.) that are required for you to get to market? If so, what is your plan to complete it?	Yes. Because SPARK's Intelligent Energy Hub will interface with electrical infrastructure, it is expected to require: - UL 916 (Energy Management Equipment) - Possibly UL 67 or UL 508, depending on final panel integration Plan: - Engage with a compliance consultant during hardware prototyping (target: Q1 2026) - Design the enclosure and electronics to meet the appropriate standard - Allocate \$50,000–\$75,000 for certification testing with an NRTL (e.g., UL, Intertek) - Complete certification by Q3/Q4 2026
If your product is software-based, how has it been tested to know that it performs as expected and/or creates the intended results?	- SPARK's core software (AI coordination engine and preference-based control layer) has been tested using real-world commercial building load data. - Simulations confirm the ability to shift, shed, and coordinate loads while honoring user-defined comfort and operational intents. - Performance benchmarks show potential 15–30% energy cost savings in simulation scenarios.

	- Current testing includes override logic, prediction modeling, and user control response simulations. - Results validate both the functional architecture and business logic behind SPARK's approach.
What is your strategy to do further technical validation of your technology and product? Who can do such validation and what will it cost?	Next steps: 1. Complete working prototype by Q4 2025. 2. Build and test initial hardware MVP in partnership with Labs and contract engineering firm (Q1 2026) 3. Run live pilots in commercial buildings (100k–200k sq ft) starting early-2026 4. Engage third-party evaluation partners (e.g., NREL, academic labs, or real estate tech partners) for technical validation and performance verification Estimated costs: - Hardware prototype: \$500,000 Pilot deployment + monitoring: \$250,000–\$300,000 across 2–3 sites - Certification & testing: \$50,000–\$75,000

If available, and relevant, provide **written technology-related testimonials** and concise evidence from credible sources that your product design can work. If you're ready, or nearly ready, for manufacture at scale, provide clear evidence or an industry third-party testimonial that there are no residual technical issues to be resolved and that you have credibly cost the manufacture. **Please upload these as a PDF (one to three pages maximum).**